Project Progress Report

Report No. 004

Report Date	Project Designation	Report By
01/09/2026	**R.AM.E. R :** Restoration of Autonomous Mobile Educational Robot. **Unit 1**	Afini Chrsitopher Allaputa

Project Description

To troubleshoot, repair and create suitable maintenance documentation of the educational Robotics Platform.

Project Task Plan

Task No.	Task Description	Status	Remarks
1	Identify and document all components in the Robot	**Completed**	Components identified
2	Test individual robot components, note functional and faulty components	**Completed**	
3	Repair or replace faulty components of university spare parts	**Completed**	
4	Integrate all I/O components and power to the Arduino and confirm input and output signals.	**Completed**	
5	Document Arduino pin out connections	**Ongoing**	
6	Confirm serial communication between Raspi and Arduino	**Completed**	Communication established
7	Document any modifications made to the system to accommodate the absence of certain I/O devices, including any functionality changes implemented to make effective use of the available hardware.	**Ongoing**	

Activity Report

Task No.	Description	Comments	% Completion
1	Successfully identified and documented 24 components on the Robot.	Some components would require dismantling the robot to identify.	100
2	Confirmed 3 faulty components	1. Replaced rasp-camera, new camera functional. 2. Replaced DC-DC voltage converter module, new module is functional 3. Replaced faulty motor, new motor is functional. 4. 802.11n Wi-Fi dongle is faulty. Currently testing headless	90

		connection between raspi and pc using LAN cable. 5. Testing custom PCB - ongoing	
3	Establish connection between Raspberry pi and Arduino	SSH connection between Raspi and Pc is currently on LAN ethernet. Serial connection between Arduino and Raspberry pi is currently over USB	100
4	Created a new raspberry pi API to interface with the Arduino firmware. Arduino firmware transmits sensor data and the raspberry pi receives and processes		100

Look Ahead

1). Change SSH connection from PC to raspberry pi to wireless.

2). Change serial connection of Arduino- Raspi to use UART pins